Controlled Systems Lab · Offline renderer

Saved stage projections

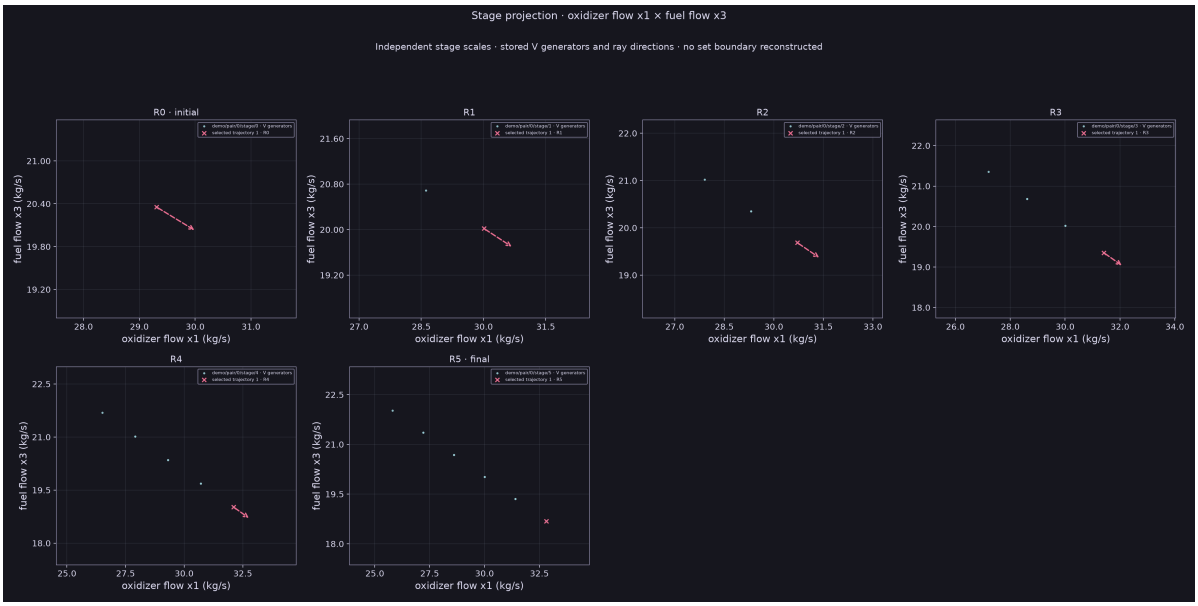
Validated batch `csl.stage-projection-batch/1`; route `demo.affine-box-reachability`; problem `fuel-four-state-box-control-demo`.

Output directory: `this bundle directory`

Each panel is a saved stage outcome using independent scales. Stored V generators are shown as markers; no boundary is reconstructed. Trajectory overlays are selected histories, separate from reachable-set geometry and evidence.

Stage projection · oxidizer flow x1 × fuel flow x3

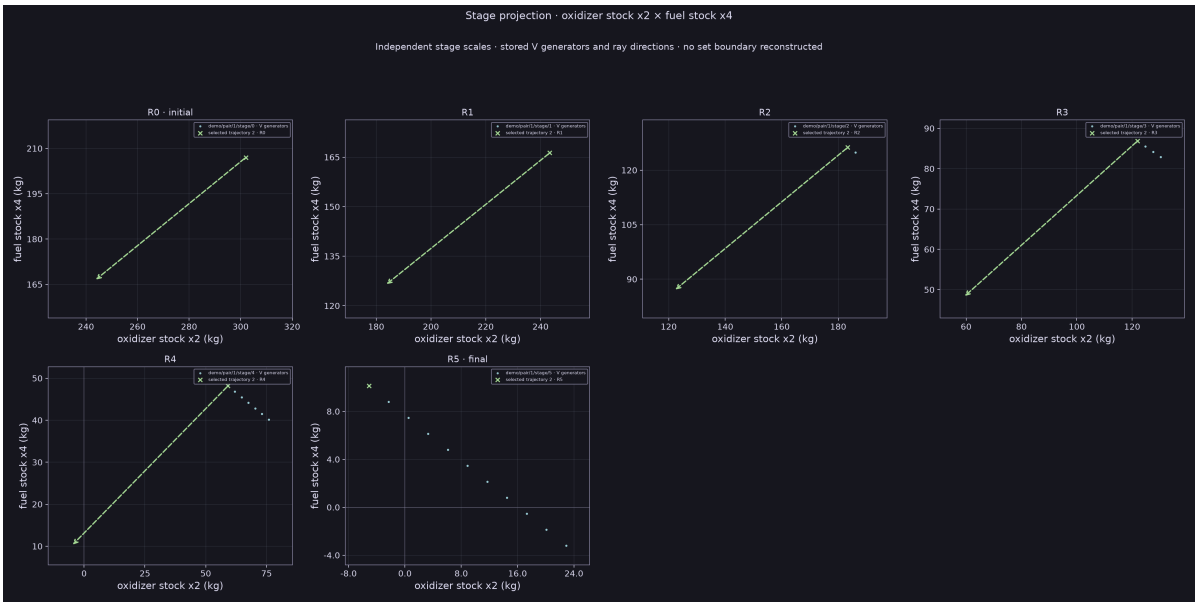
Independent scales are used for each stage. R0 · initial: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R0']; evidence scopes set_relation. component demo/pair/0/stage/0: point, 1 stored V generators, 0 rays. selected trajectory 1 at R0 · initial; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R1: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R1']; evidence scopes set_relation. component demo/pair/0/stage/1: segment, 2 stored V generators, 0 rays. selected trajectory 1 at R1; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R2: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R2']; evidence scopes set_relation. component demo/pair/0/stage/2: segment, 4 stored V generators, 0 rays. selected trajectory 1 at R2; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R3: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R3']; evidence scopes set_relation. component demo/pair/0/stage/3: segment, 8 stored V generators, 0 rays. selected trajectory 1 at R3; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R4: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R4']; evidence scopes set_relation. component demo/pair/0/stage/4: polygon, 16 stored V generators, 0 rays. selected trajectory 1 at R4; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R5 · final: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R5']; evidence scopes set_relation. component demo/pair/0/stage/5: polygon, 32 stored V generators, 0 rays. selected trajectory 1 at R5 · final; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry. The figure plots saved V generators and ray directions. It does not reconstruct set boundaries or perform geometry conversion.



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Stage projection · oxidizer stock x2 × fuel stock x4

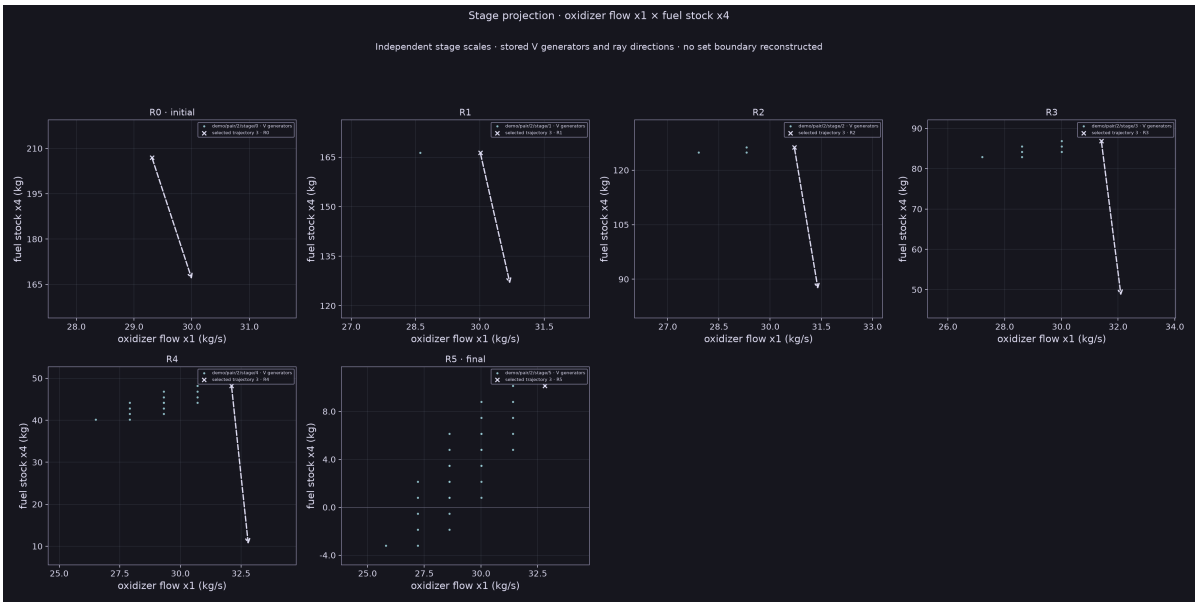
Independent scales are used for each stage. R0 · initial: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R0']; evidence scopes set_relation. component demo/pair/1/stage/0: point, 1 stored V generators, 0 rays. selected trajectory 2 at R0 · initial; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R1: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R1']; evidence scopes set_relation. component demo/pair/1/stage/1: point, 1 stored V generators, 0 rays. selected trajectory 2 at R1; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R2: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R2']; evidence scopes set_relation. component demo/pair/1/stage/2: segment, 4 stored V generators, 0 rays. selected trajectory 2 at R2; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R3: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R3']; evidence scopes set_relation. component demo/pair/1/stage/3: polygon, 8 stored V generators, 0 rays. selected trajectory 2 at R3; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R4: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R4']; evidence scopes set_relation. component demo/pair/1/stage/4: polygon, 16 stored V generators, 0 rays. selected trajectory 2 at R4; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R5 · final: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R5']; evidence scopes set_relation. component demo/pair/1/stage/5: polygon, 32 stored V generators, 0 rays. selected trajectory 2 at R5 · final; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry. The figure plots saved V generators and ray directions. It does not reconstruct set boundaries or perform geometry conversion.



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Stage projection · oxidizer flow x1 × fuel stock x4

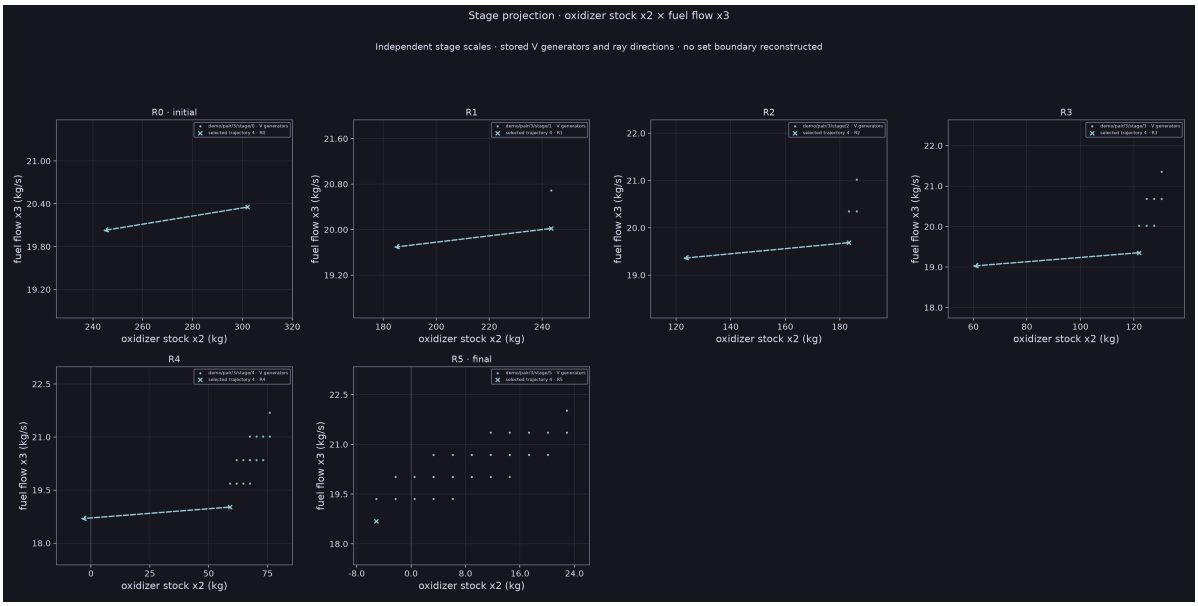
Independent scales are used for each stage. R0 · initial: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R0']; evidence scopes set_relation. component demo/pair/2/stage/0: point, 1 stored V generators, 0 rays. selected trajectory 3 at R0 · initial; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R1: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R1']; evidence scopes set_relation. component demo/pair/2/stage/1: segment, 2 stored V generators, 0 rays. selected trajectory 3 at R1; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R2: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R2']; evidence scopes set_relation. component demo/pair/2/stage/2: polygon, 4 stored V generators, 0 rays. selected trajectory 3 at R2; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R3: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R3']; evidence scopes set_relation. component demo/pair/2/stage/3: polygon, 8 stored V generators, 0 rays. selected trajectory 3 at R3; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R4: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R4']; evidence scopes set_relation. component demo/pair/2/stage/4: polygon, 16 stored V generators, 0 rays. selected trajectory 3 at R4; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R5 · final: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R5']; evidence scopes set_relation. component demo/pair/2/stage/5: polygon, 32 stored V generators, 0 rays. selected trajectory 3 at R5 · final; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry. The figure plots saved V generators and ray directions. It does not reconstruct set boundaries or perform geometry conversion.



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Stage projection · oxidizer stock x2 × fuel flow x3

Independent scales are used for each stage. R0 · initial: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R0']; evidence scopes set_relation. component demo/pair/3/stage/0: point, 1 stored V generators, 0 rays. selected trajectory 4 at R0 · initial; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R1: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R1']; evidence scopes set_relation. component demo/pair/3/stage/1: segment, 2 stored V generators, 0 rays. selected trajectory 4 at R1; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R2: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R2']; evidence scopes set_relation. component demo/pair/3/stage/2: polygon, 4 stored V generators, 0 rays. selected trajectory 4 at R2; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R3: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R3']; evidence scopes set_relation. component demo/pair/3/stage/3: polygon, 8 stored V generators, 0 rays. selected trajectory 4 at R3; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R4: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R4']; evidence scopes set_relation. component demo/pair/3/stage/4: polygon, 16 stored V generators, 0 rays. selected trajectory 4 at R4; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry; R5 · final: success / equivalent. relation equivalent; assurance numerical; provider csl.demo.affine-box-coordinate-selection version 1.0.0; source sets ['demo-state-set-R5']; evidence scopes set_relation. component demo/pair/3/stage/5: polygon, 32 stored V generators, 0 rays. selected trajectory 4 at R5 · final; evidence: illustrative fixed control u=1 at every stage; not optimized; separate from set geometry. The figure plots saved V generators and ray directions. It does not reconstruct set boundaries or perform geometry conversion.



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